

Title:

Analyzing Adverse Drug Events Using OpenFDA Data

Description:

For my capstone project, I have selected the **OpenFDA Adverse Event Reporting System (FAERS)** dataset. This publicly available dataset contains detailed reports of negative drug reactions collected by the FDA, reported by healthcare professionals, consumers, and pharmaceutical companies.

The objective of this project is to design and build a **scalable cloud-based data pipeline** that can ingest, process, and analyze these adverse drug events. The pipeline will extract meaningful insights such as:

- Identification of drugs with the highest number of adverse events
- Common types of adverse reactions per drug
- Patterns across patient demographics (age, gender, etc.)

This project is aligned with real-world data engineering tasks involving **ETL processes**, **semi-structured data (JSON)**, and **cloud deployment**.

Additionally, this topic bridges my software development background with healthcare-related data engineering challenges, helping me transition smoothly into the field.